ALAN LAMB

Billings, MT | Open to relocation | (406) 598-4073 | alandaniellamb2@gmail.com | linkedin.com/in/alanlamb-econ | lambcast.net

TECHNICAL SKILLS

Econometrics & Statistical Analysis: Regression modeling (OLS, fixed effects, DiD, synthetic control), panel data methods, wild cluster bootstrap, structural break testing, welfare analysis, causal inference

Software & Tools: Stata (advanced), Python (intermediate), R (intermediate), Git, Microsoft Office Suite (Word, Excel, PowerPoint)

Data: EIA Form 930 API, LBNL interconnection queue data, multi-source merging (Census, BEA, NOAA), large dataset cleaning and validation

EDUCATION

M.S. Applied Economics (STEM-designated) | University of Maryland | Expected December 2026

Coursework: Empirical Analysis I-III, Environmental Economics, Microeconomic Analysis, Economic Applications of R Programming

Master of Public Administration | University of Montana | May 2013

B.S. Political Science | Montana State University Billings | April 2010

APPLIED RESEARCH

AI Infrastructure and Regional Electricity Demand: Evidence from U.S. Interconnection Queues

Independent Research | University of Maryland | March 2026 | SSRN: ssrn.com/abstract=6446446 | Dashboard: bit.ly/4siJ0VL

- Estimated the causal contribution of AI data center investment to regional electricity demand divergence across ERCOT, PJM, and MISO using 184,000+ hourly EIA observations and 16,000+ LBNL interconnection queue filings from 2019 to 2025
- Designed a four-layer identification strategy where each method fails differently: panel regression establishes a conditional association under common trends; synthetic control builds a counterfactual from low-exposure donors and establishes a lower bound; difference-in-differences with clean controls removes contamination bias and produces an upper bound; and narrative validation ties the statistical gap to documented facility energization dates. Convergence across all four layers, despite their different failure modes, is the evidence
- Headline result: 34.8 index-point divergence in ERCOT minimum hourly demand above a synthetic counterfactual; zero of three placebos exceed the gap; robust across all ten panel specifications and three donor pool configurations
- Built a scenario-based forecasting pipeline that casts a net rather than a point estimate: an ARIMA baseline captures the historical time-series structure; queue-augmented models incorporate the investment pipeline signal; and historical completion rates (ERCOT 33.9%, PJM 19.7%, MISO 17.8%) bound conservative and trend scenarios through 2027, quantifying what better load-side data disclosure would reduce
- Produced an interactive published dashboard in Hex visualizing zone-level ERCOT load growth, large-load queue filings by region, project completion rates, and forward demand projections through 2027
- Documented the absence of public large-load interconnection queue data as a policy finding; argued for FERC disclosure requirements modeled on Georgia Power and Texas SB-6 precedents

Bearing | AI Daily Intelligence PWA (bearing.fyi)

Independent Software Development | 2025 to Present

- Designed and shipped a full-stack production web app as sole developer, building auth, payments, AI integration, and cross-device sync from zero with no external engineering support, demonstrating end-to-end ownership and independent delivery on a complex technical project

PROFESSIONAL EXPERIENCE

Montana Brewing Company & Hooligans Sports Bar | Billings, MT March 2010 - February 2024

Front of House Manager - June 2019 - February 2024

Bartender - October 2011 - June 2019

Security / Barback - March 2010 - October 2011

- Progressed from entry-level to managing FOH operations across a high-volume, dual-concept venue; directed teams of 20 to 40 per shift, oversaw hiring and onboarding, and handled escalations and service recovery across a 14-year tenure